

Aptitude Systems

Beacon Scoring Engine

Six-Dimensional Sub-Metric Implementation Specification

Aptitude Systems — Internal Documentation

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*Companion to: Aptitudinal Alignment Quantum Manual v0.1
Volumes I–VI, Institutional Fidelity Field Theory*

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Architecture Independence Declaration

This specification governs the Beacon Analytics Service — the institutional scoring pipeline of Aptitude Systems. Before any dimensional specification is read, the following invariant must be understood and enforced throughout implementation:

INVARIANT B-0: INSTITUTIONAL INDEPENDENCE

The Beacon scoring system is architecturally independent of individual user activity. Aptitude Scores are derived from public institutional records, regulatory filings, court decisions, and agency-generated data. Individual incident submissions from the Doc 1 pipeline are a *supplementary signal source* — one channel among many — and their absence does not degrade institutional score validity. An institution cannot suppress, dilute, or game its Aptitude Score by discouraging individual filings. **The public record is the primary and sufficient evidentiary basis.**

This independence is enforced at the data layer: the `behavioral_quanta` table accepts a `source_type` field. Only records with:

```
source_type IN (
  'public_record', 'regulatory_filing', 'court_decision',
  'agency_report', 'cfpb_complaint', 'ftc_action',
  'inspector_general'
)
```

are eligible for institutional score computation. Records with `source_type = 'individual_submission'` are tagged as supplementary and weighted at $\leq 5\%$ of any dimensional score unless corroborated by a public record.

Composite Scoring Formula

Aptitude Score (A)

The Aptitude Score is a weighted linear combination of the six dimensional scores:

$$A = \sum_{i=1}^6 w_i \cdot D_i \quad \text{where } D_i \in \{F, C, T, E, Q, Ac\}$$

$$\text{subject to: } \sum_{i=1}^6 w_i = 1.0, \quad w_i > 0 \forall i$$

Default AHP-derived weights, configurable per institution class:

Dimension	Symbol	Default Weight	Rationale
Fidelity	F	$w_1 = 0.22$	Legal mandate adherence is foundational
Consistency	C	$w_2 = 0.18$	Variance in outcomes signals systemic bias
Transparency	T	$w_3 = 0.17$	Public record access is a legal right
Efficiency	E	$w_4 = 0.15$	Resource stewardship proxies competence
Equity	Q	$w_5 = 0.16$	Demographic fairness is a constitutional floor
Accountability	Ac	$w_6 = 0.12$	Self-correction capacity determines trajectory

Min-Max Normalization

All sub-metric raw values are normalized before dimensional score computation:

$$D_{\text{norm}} = \frac{D_{\text{raw}} - D_{\text{min}}}{D_{\text{max}} - D_{\text{min}}}$$

Bounds are configurable per institution class (see Section 10). Values are clamped to $[0, 1]$ — no extrapolation permitted. Normalized values are stored in `behavioral_quanta.magnitude`.

Classification Tiers

Tier	Score Range	Label	System Action
I	0.85 – 1.00	High Alignment	Green badge; no flags; eligible for public trust designation
II	0.70 – 0.84	Adequate	Yellow badge; flag weakest dimension for monitoring
III	0.50 – 0.69	At Risk	Orange badge; trigger audit workflow; notify stakeholder layer
IV	0.30 – 0.49	Low Alignment	Red badge; flag to oversight bodies; publish dimension breakdown
V	0.00 – 0.29	Critical Failure	Black badge; escalate to regulatory referral queue; public disclosure

Dimension 1: Fidelity (Φ_F)

FIDELITY — Measures whether an institution adheres to its legal mandate, enabling statutes, and constitutional obligations. The fidelity field collapses to zero when an institution acts outside statutory authority, systematically ignores regulatory obligations, or applies law selectively.

Dimensional Formula

$$F = w_1 \cdot \text{LMCR} + w_2 \cdot \text{PAS} + w_3 \cdot \text{RVR} + w_4 \cdot (1 - \text{SOF})$$

where LMCR = Legal Mandate Compliance Rate; PAS = Policy Adherence Score; RVR = Regulatory Violation Reversal Rate; SOF = Statutory Override Frequency (inverted).

Sub-Metric Specification

Code	Name	Definition	Source	Range	Wt.
LMCR	Legal Mandate Compliance Rate	Ratio of actions complying with enabling statute to total actions. Computed from regulatory findings, court orders, and IG reports.	CFPB exams; OCC findings; State AG filings; Court orders	[0, 1]	0.30
PAS	Policy Adherence Score	Ratio of internal policy decisions aligning with promulgated regulations. Derived from enforcement action records.	FTC actions; CFPB consent orders; Regulatory correspondence	[0, 1]	0.25
RVR	Regulatory Violation Reversal Rate	Fraction of identified violations corrected before external enforcement. Self-correction elevates fidelity; forced correction does not.	IG reports; Voluntary corrective action filings; Audit records	[0, 1]	0.25
SOF	Statutory Override Frequency	Rate at which institution actions required courts or regulators to enforce statutory limits. Inverted in formula.	Federal/state court dockets; Regulatory enforcement database	[0, 1]	0.20

Agent Specifications

AGENT: Fidelity Signal Parser — Mandate Compliance Extraction

Inputs: Raw regulatory filings, court orders, IG reports, consent orders (JSON/PDF)

Outputs: Structured fidelity quanta with LMCR/PAS/RVR/SOF magnitude per action

Invariants: ZERO tolerance for uncited compliance claims. Every quantum must carry `source_id`, `document_date`, and `jurisdiction_code`. Agent MUST flag ambiguous compliance characterizations for human review — it does not resolve them. Rejects any document older than 7 years unless it constitutes ongoing pattern evidence.

AGENT: Fidelity Field Synthesizer — Dimensional Score Computation

Inputs: Normalized fidelity quanta for target institution and time window

Outputs: F score $\in [0, 1]$ with confidence interval, quantum count, source breakdown, and outlier flags

Invariants: Minimum 12 quanta required for score computation. If quantum count < 12 , emit `INSUFFICIENT_DATA` status — do not estimate. Outliers ($|x - \mu| > 2\sigma$) are retained but flagged. Score must be reproducible byte-for-byte from the quantum set.

Dimensional Invariants

- F score requires a minimum of 12 behavioral quanta from at least 3 distinct source types.
- Consent orders count as negative compliance evidence regardless of whether the institution "agreed" to them.
- Self-reported compliance documentation receives a 0.50 credibility discount unless corroborated by independent regulatory finding.
- A single court finding of *ultra vires* action triggers an automatic LMCR floor of 0.20 for the affected period regardless of other quanta.

Dimension 2: Consistency (Φ_C)

CONSISTENCY — Measures variance in institutional outcomes across substantively similar cases. Inconsistency is the statistical signature of bias, favoritism, and arbitrary decision-making. Consistency measures whether like cases are treated alike — not whether outcomes are correct.

Dimensional Formula

$$C = w_1 \cdot (1 - \text{OVI}) + w_2 \cdot \text{PAR} + w_3 \cdot (1 - \text{CDS}) + w_4 \cdot (1 - \text{DRR})$$

Sub-Metric Specification

Code	Name	Definition	Source	Range	Wt.
OVI	Outcome Variance Index	Statistical variance in outcomes across substantively similar cases scored by the Legal Classifier Agent. <i>Inverted in formula.</i>	Case outcome databases; FOIA decision records; Public dockets	[0, 1]	0.30
PAR	Precedent Adherence Rate	Rate at which current decisions align with the institution's own prior decisions in analogous cases.	Administrative law records; Internal guidance; Published decisions	[0, 1]	0.25
CDS	Cross-Case Disparity Score	Magnitude of outcome differences between demographically distinct but legally identical case populations. <i>Inverted.</i>	HMDA data; CFPB demographic reports; Fair lending exams; EEOC filings	[0, 1]	0.25
DRR	Decision Reversal Rate	Rate at which institution decisions are overturned on appeal or administrative review. <i>Inverted.</i>	Federal/state appellate records; Administrative review outcomes	[0, 1]	0.20

Agent Specifications

AGENT: Case Similarity Engine — Substantive Equivalence Classification

Inputs: Raw case records with outcome data, demographic indicators, and procedural history

Outputs: Case clusters with similarity scores; OVI and CDS raw values per cluster

Invariants: Similarity scoring uses semantic embedding (VECTOR(1536)) + struc-

tured attribute matching. Demographic attributes are isolated variables — they must NOT appear as similarity factors. Engine must expose its similarity rationale for every cluster. Minimum cluster size = 30 cases before OVI is computed.

AGENT: Consistency Auditor — Variance and Reversal Computation

Inputs: Case clusters, precedent database, appellate outcome records

Outputs: C score $\in [0, 1]$ with per-sub-metric breakdown, cluster report, and disparity flags

Invariants: Any sub-metric below 0.30 triggers `CONSISTENCY_ALERT` appended to the score record. Reversal data must be sourced from appellate court directly — institution self-reporting of reversal rates is not accepted. CDS computation must be independently reproduced for any Tier IV or V classification.

Dimensional Invariants

- OVI is computed at the case-cluster level. A single high-variance cluster is published separately without suppressing the overall score.
- Demographic disparity above 15% in any protected class triggers an `EQUITY CROSS-FLAG` regardless of overall C score.
- Institutions under court-ordered desegregation or consent decrees carry those records as permanent low-consistency markers until a 5-year clean record is established.
- Selective retroactive application of policy changes is penalized; uniform prospective changes are not.

Dimension 3: Transparency (Φ_T)

TRANSPARENCY — Measures whether the institution makes its records, decisions, and procedures accessible. Transparency is not a courtesy — for most public institutions it is a legal requirement. The transparency field penalizes stonewalling, delay, redaction abuse, and procedural opacity with the same weight as outright violation.

Dimensional Formula

$$T = w_1 \cdot \text{RDR} + w_2 \cdot \text{RTI} + w_3 \cdot \text{FCR} + w_4 \cdot \text{PAX}$$

Sub-Metric Specification

Code	Name	Definition	Source	Range	Wt.
RDR	Record Disclosure Rate	Fraction of requested records fully disclosed without substantively contested redactions. Partial disclosures count proportionally.	FOIA.gov; State public records databases; Disclosure litigation records	[0, 1]	0.28
RTI	Response Time Index	min(deadline/median_actual, Faster than statutory deadline = 1.0; slower degrades proportionally.	FOIA logs cross-validated with requester data; Court dockets	[0, 1]	0.25
FCR	FOIA/Public Records Compliance Rate	Rate of requests fulfilled without requiring administrative appeal or litigation. Legally compelled disclosures are counted as non-compliant.	DOJ OIP FOIA statistics; State AG reports; Litigation outcomes	[0, 1]	0.27
PAX	Proactive Access Index	Rate at which institution voluntarily publishes decision rationales, policy changes, and statistical reports without being requested. Benchmarked against peer class.	Agency public portals; Federal Register; State rule publication logs	[0, 1]	0.20

Agent Specifications

AGENT: Transparency Crawler — Public Record Availability Mapping

Inputs: Agency identifiers, known public portals, FOIA log databases, court docket APIs

Outputs: Transparency quanta per agency with RDR/RTI/FCR/PAX raw values and citation chain

Invariants: Agent must verify record availability from a non-institutional network path. Availability that requires login, payment, or special permission is treated as restricted. Re-crawl every 30 days. Any FOIA litigation loss automatically sets FCR quantum = 0.0 for the contested record category for 24 months.

AGENT: Transparency Adjudicator — Redaction and Delay Scoring

Inputs: Disclosure records, response timelines, redaction justification codes

Outputs: T score $\in [0, 1]$; redaction abuse flag; delay pattern analysis; proactive publication inventory

Invariants: Deliberative process privilege and FOIA Exemption 5 are the most abused exemptions — any invocation triggers heightened scrutiny. Agent must crosscheck claimed exemption against successful appeal rates for that exemption at the same agency. Pattern of exemption abuse ($> 25\%$ appeal overturn rate) generates a REDACTION_ABUSE flag that reduces RDR by 0.15.

Dimensional Invariants

- Response time is measured from the date of institutional *receipt*, not date of acknowledgment — the clock cannot be reset by delaying acknowledgment.
- Institutions requiring in-person or physical mail submission only receive an automatic PAX deduction of 0.10.
- Sealed settlement agreements containing public interest matters are treated as non-disclosures for RDR purposes.
- An institution held in contempt for disclosure failure within the last 5 years cannot score above $T = 0.40$ regardless of subsequent performance.

Dimension 4: Efficiency (Φ_E)

EFFICIENCY — Measures how effectively an institution converts public resources — funding, staffing, authority — into mandated outcomes. Efficiency is not austerity: an institution that is underfunded but maximizes what it has scores high. Efficiency is the closest proxy to institutional competence.

Dimensional Formula

$$E = w_1 \cdot \text{CRR} + w_2 \cdot \text{PTS} + w_3 \cdot \text{RUI} + w_4 \cdot \text{BCR}$$

Sub-Metric Specification

Code	Name	Definition	Source	Range	Wt.
CRR	Case Resolution Rate	Fraction of cases resolved within the applicable statutory or regulatory time limit. Administrative closures without documented outcome do <i>not</i> count.	Agency performance reports; Congressional testimony; GAO/IG audits	[0, 1]	0.28
PTS	Processing Time Score	$\min(\text{benchmark_median}/\text{agency}, \text{benchmark})$. Normalized against peer-class institutional benchmark and statutory limit.	Annual reports; FOIA case management logs; Regulatory performance standards	[0, 1]	0.25
RUI	Resource Utilization Index	Ratio of measurable mandate-advancing outcomes to annual operating budget. <i>Net of litigation costs</i> . Normalized against peer class.	Agency budget justifications; Congressional appropriations; Published performance metrics	[0, 1]	0.22
BCR	Backlog Clearance Rate	Year-over-year backlog trajectory. Growing < 0.50; stable = 0.50; reducing > 0.50.	Annual reports; IG backlog findings; Congressional oversight testimony	[0, 1]	0.25

Agent Specifications

AGENT: Efficiency Metrics Extractor — Performance Data Ingestion

Inputs: Agency performance reports, budget justifications, GAO/IG audits, congressional testimony

Outputs: Efficiency quanta with CRR/PTS/RUI/BCR values, fiscal year tagged, peer-class benchmarked

Invariants: Peer-class benchmarking is MANDATORY. The agent maintains a peer-class registry keyed by `institution_type` × `budget_tier` × `jurisdiction_level`. If peer class has fewer than 5 members, emit `PEER_CLASS_INSUFFICIENT` and do not score.

AGENT: Efficiency Trend Analyzer — Longitudinal Performance Assessment

Inputs: Efficiency quanta across minimum 3 fiscal years, budget change records

Outputs: E score $\in [0, 1]$; trend vector (improving/stable/degrading); resource shock flags

Invariants: Budget cuts exceeding 20% in a single year trigger a `BUDGET_SHOCK` flag applying a documented 0.10 grace adjustment to E for that year only. Institutions showing degrading E scores despite stable or increasing budgets receive a `MANAGEMENT_EFFICIENCY` flag surfaced in the public score breakdown.

Dimensional Invariants

- Administrative closures without documented outcome are flagged as `ADMINISTRATIVE_CLOSURE` and do not count toward CRR.
- Efficiency scores are always displayed with the peer-class benchmark visible to users.
- RUI is computed net of litigation costs — funds spent defending against the institution's own violations are excluded from the outcome-advancing numerator.
- Courts-ordered capacity constraints are documented as annotations — the constraint is noted but does not excuse backlogs.

Dimension 5: Equity (Φ_Q)

EQUITY — Measures whether the institution delivers its mandate fairly across demographic groups, geographic areas, and economic strata. Equity is the constitutional floor of institutional legitimacy. An institution that scores perfectly on every other dimension but systematically disadvantages protected classes fails the public trust at the most fundamental level.

Dimensional Formula

$$Q = w_1 \cdot (1 - \text{DDI}) + w_2 \cdot \text{GAS} + w_3 \cdot \text{PCOR} + w_4 \cdot \text{ASR}$$

Sub-Metric Specification

Code	Name	Definition	Source	Range	Wt.
DDI	Demographic Disparity Index	Composite index of outcome rate differences across race, gender, age, disability, and national origin for substantively similar cases. <i>Inverted in formula.</i>	HMDA; EEOC charge data; CFPB fair lending exams; Disparate impact litigation	[0, 1]	0.30
GAS	Geographic Access Score	Service availability and outcome quality across rural, suburban, and urban geographies. Includes physical, digital, and language access.	USPS ZIP-level data; Agency field office maps; Language access compliance reports	[0, 1]	0.23
PCOR	Protected Class Outcome Rate	Ratio of favorable outcomes for protected class members to the expected rate at parity, controlling for legally relevant variables. Score = 1.0 at parity.	Civil rights investigation outcomes; Fair lending exam results; Title VI/VII enforcement	[0, 1]	0.27
ASR	Appeals Success Rate (Protected Class)	Rate at which protected-class complainants who challenge adverse decisions prevail on appeal. High rates indicate systematic erroneous adverse decisions.	Federal/state appellate records; Administrative review outcomes; Civil rights findings	[0, 1]	0.20

Agent Specifications

AGENT: Equity Signal Extractor — Demographic Disparity Detection

Inputs: Case outcome records with demographic indicators, geographic data, HMDA/EEOC/CFPB filings

Outputs: DDI and PCOR values per demographic dimension; disparity alerts; statistical significance flags

Invariants: Minimum $n = 100$ per demographic cell before statistical claims are made. Cells below $n = 100$ are excluded and documented as `INSUFFICIENT_SAMPLE`. Agent NEVER imputes demographic data. Any disparity reaching significance at $p < 0.05$ triggers `EQUITY_ALERT` regardless of DDI magnitude.

AGENT: Equity Field Enforcer — Score Computation with Hard Floors

Inputs: Equity quanta, disparity alerts, litigation records, consent decree status

Outputs: Q score $\in [0, 1]$; mandatory flag set; hard-floor enforcement log; public disclosure summary

Hard Floors (override weighted average — cannot be bypassed):

- Finding of intentional discrimination within 10 years $\Rightarrow Q_{\min} = 0.15$
- Active consent decree for civil rights violation $\Rightarrow Q_{\min} = 0.25$
- Pattern-or-practice finding $\Rightarrow Q_{\min} = 0.20$

All floors are documented in the score record. Hard floors cannot be overridden by composite score computation.

Dimensional Invariants

- Equity is the only dimension with mandatory hard floors that override the weighted average.
- An institution may submit a legally-relevant variable model to justify disparity — that model must be validated by the Signal Harmonizer Agent before it is accepted.
- Geographic access scoring includes digital-only alternatives only if 90%+ of the service population has documented reliable broadband access.
- Language access failures are treated as geographic access failures — a physically present but linguistically inaccessible office scores no higher than an absent one.
- A Tier V equity score triggers a mandatory regulatory referral workflow regardless of composite Aptitude Score.

Dimension 6: Accountability (Φ_{Ac})

ACCOUNTABILITY — Measures the institution’s capacity and willingness to identify its own failures, correct them, and institutionalize reforms that prevent recurrence. Accountability is the temporal dimension — it captures trajectory, not just state. It is the dimension most predictive of future behavior.

Dimensional Formula

$$Ac = w_1 \cdot SCR + w_2 \cdot AC + w_3 \cdot CRR_{comp} + w_4 \cdot RAR$$

where SCR = Self-Correction Rate; AC = Audit Compliance; CRR_{comp} = Complaint Resolution Rate (quality measure, distinct from Efficiency CRR); RAR = Remediation Action Rate.

Sub-Metric Specification

Code	Name	Definition	Source	Range	Wt.
SCR	Self-Correction Rate	Rate at which institution identifies and corrects errors through its own review processes before external enforcement, litigation, or complaint compels action. Proactive correction receives 1.5× multiplier.	Voluntary corrective action filings; Annual report disclosures; IG management responses	[0, 1]	0.28
AC	Audit Compliance	Rate of compliance with internal audit recommendations, regulatory examination findings, and IG recommendations within the mandated response window. Tracks <i>implementation</i> , not acknowledgment.	IG semiannual reports; GAO follow-up audits; Regulatory examination tracking	[0, 1]	0.25
CRR_c	Complaint Resolution Rate	Rate of consumer/public complaints resolved to documented satisfaction within the institution’s stated timeline. Audited by 10% random sampling of “resolved” complaints.	CFPB complaint database; BBB records; State AG databases; Agency complaint logs (FOIA)	[0, 1]	0.22

Code	Name	Definition	Source	Range	Wt.
RAR	Remediation Action Rate	Fraction of committed remediation actions implemented on the committed timeline. Delayed implementation scores as partial credit proportional to delay magnitude.	Consent order monitoring reports; Court-appointed monitor reports; IG follow-up findings	[0, 1]	0.25

Agent Specifications

AGENT: Self-Audit Intelligence Agent — Correction Origin Classification

Inputs: Corrective action records, enforcement timelines, complaint databases, audit response logs

Outputs: SCR and RAR values with correction-origin classification (self-initiated vs. externally compelled) and timeline analysis

Invariants: Classification is binary: self-initiated (SCR credit) or externally compelled (no SCR credit). A correction triggered by a complaint that the institution received but had not yet acted on for > 30 days is classified as externally compelled. Agent must maintain a correction provenance chain for every SCR quantum — the origin must be independently verifiable.

AGENT: Accountability Enforcement Agent — Commitment vs. Performance Tracking

Inputs: Consent order commitments, audit recommendation logs, remediation schedules, IG monitoring reports

Outputs: Ac score $\in [0, 1]$; commitment fulfillment ledger; chronic non-compliance flags; trajectory vector

Invariants: CHRONIC_ACCOUNTABILITY_FAILURE flag is set if $AC < 0.40$ for 3 consecutive scoring periods. Flag persists for 5 years and is published in the public score breakdown. Remediation promises without documented implementation score as zero. Resolution fraud detection: 10% random sample of "resolved" complaints is traced; fabricated resolutions generate RESOLUTION_FRAUD flag.

Dimensional Invariants

- SCR receives a $1.5\times$ multiplier for corrections made proactively before any complaint or audit raised the issue.
- A pattern of recurring identical violations (same type in 3+ consecutive periods) automatically caps Ac at 0.35.

- Institutions in receivership, under mandatory monitor, or under a corporate integrity agreement are scored against their compliance schedule, not voluntary benchmarks.
- Institutions that mark complaints resolved without documented resolution receive a `RESOLUTION_FRAUD` flag that triggers an audit of all CRR scores for the preceding 24 months.

Credit Reporting Agency (CRA) Institutional Mapping

CRA Class Definition

Field	Value
Institution Class	Nationwide Consumer Reporting Agency (NCRA)
Regulatory Framework	FCRA, 15 U.S.C. § 1681 <i>et seq.</i> ; Regulation V (12 C.F.R. Part 1022)
Primary Regulator	Consumer Financial Protection Bureau (CFPB)
Secondary Regulators	FTC; State financial regulators; State AG offices
Peer Class Size	3 NCRA (primary); $\approx$ 30 specialty CRAs (secondary cohort)
Scoring Frequency	Quarterly dimensional; Annual composite Aptitude Score
Hard Floors	All six dimensional hard floors apply

CRA-Specific Data Requirements — Complete Inventory

A. Regulatory and Legal Data

- CFPB supervisory examination reports (obtain via consent order summaries and FOIA)
- CFPB enforcement actions, consent orders, and civil investigative demands
- FTC enforcement actions and complaints against CRAs (public docket)
- State AG enforcement actions — all 50 states + DC
- FCRA civil litigation outcomes — PACER, CourtListener
- Class action settlements and consent decrees — amounts, injunctive terms, compliance schedules
- Court-appointed monitor reports
- Federal and state appellate decisions involving CRA defendants
- Congressional testimony by CRA executives (HFSC, Senate Banking Committee)
- GAO reports on credit reporting accuracy and CRA practices
- CFPB Supervisory Highlights mentioning credit reporting industry
- Inspector General reports on CFPB CRA supervision

B. Consumer Complaint Data

- CFPB Consumer Complaint Database — all CRA complaints (downloadable, daily updated)
- FTC Consumer Sentinel Network — CRA-related complaints
- State AG consumer complaint databases (FOIA where not public)
- BBB complaint records for each CRA entity

- Complaint resolution rates and timelines (computed from database timestamps)
- Re-opened complaint rates (failed first resolution signal)
- Complaint category distribution: dispute inaccuracy, identity theft, improper use, debt collector, fraud alert

C. FCRA Compliance Data (Accuracy and Dispute)

- Dispute volume and resolution timelines per CRA (CFPB data)
- Reinvestigation outcomes: verified, modified, deleted, unresolvable
- Re-dispute rates (same item disputed multiple times — accuracy failure signal)
- Maximum investigation period compliance (30/45-day statutory limits under § 1681i)
- Furnisher notification rates and timelines
- Block implementation rates (identity theft blocks under § 1681c-2)
- Security freeze compliance — placement, lift, removal timelines
- Annual credit report delivery compliance (AnnualCreditReport.com)
- CFPB accuracy study data (2012 study; any subsequent studies)
- FTC accuracy study data (2013 Report to Congress and follow-ups)

D. Financial and Operational Data

- Annual reports and 10-K filings (Equifax NYSE: EFX; Experian LSE: EXPN; TransUnion NYSE: TRU)
- 10-Q quarterly filings — legal proceedings section and risk factors
- Legal contingency disclosures (litigation reserves — material dispute signal)
- Cybersecurity incident disclosures — SEC 8-K filings (material breach)
- Data breach notification records — state AG breach databases
- Equifax 2017 breach settlement compliance reports (ongoing monitoring)
- CISA cybersecurity incident reports involving CRA systems

E. Equity and Demographic Data

- CFPB fair lending examination findings for credit reporting context
- Racial and demographic credit score distribution studies (Urban Institute; Brookings; CFPB research)
- Geographic access data: states/ZIP codes with disproportionately low credit file coverage

- Thin-file and no-file population demographics (CFPB "Data Point: Credit Invisibles")
- Language access compliance: Spanish-language dispute processes and reports
- Disability access compliance: screen reader compatibility, accessible dispute processes
- Income-stratified dispute resolution success rates
- Federal Reserve studies on credit reporting and racial wealth gap

F. Transparency and Public Record Data

- CFPB rulemaking comments and CRA participation records
- Public data format availability: machine-readable schemas, API access terms
- Opt-out mechanism availability and effectiveness (FCRA § 1681b consumer rights)
- Credit freeze fee history (pre- and post-Economic Growth Act 2018 prohibition)
- Consumer disclosure request compliance (§ 1681g — full file disclosure)
- CFPB public enforcement action docket
- State-level required disclosure publication compliance

G. Accountability and Audit Data

- Consent order compliance monitoring reports
- FTC CRA audit findings (non-public → FOIA attempt; supplement with docket records)
- CFPB supervisory examination follow-up correspondence
- Self-reported accuracy improvement initiatives and measured outcomes
- Data furnisher dispute process audit results
- E-OSCAR system audit records (dispute processing system)
- Voluntary corrective action announcements and implementation evidence
- Congressional follow-up responses to oversight requests
- Recidivism tracking: same violation types across multiple enforcement periods

CRA-Specific Normalization Bounds

Bounds calibrated to the NCRA peer class, reflecting operational realities and historical performance range of the three NCRAAs specifically.

Code	Raw Unit	$D_{\min}$	$D_{\max}$	Calibration Rationale
LMCR	% actions compliant	40%	100%	No NCRA has sustained 100% compliance; 40% = worst enforcement period
OVI	σ^2 of outcomes	0.00	0.35	Variance > 0.35 indicates systematic failure
RDR	% records disclosed	30%	100%	CRAs historically contested disclosure; floor at documented nadir
RTI	deadline/actual ratio	0.10	1.00	Worst documented response = 300 days; ratio = 0.10
FCR	% without compelled action	20%	100%	Pre-2017 CFPB era baseline
CRR	% cases on time	30%	100%	Dispute resolution timeliness from CFPB database
DDI	composite disparity index	0.00	0.50	Disparity > 50% in any cell triggers hard floor
SCR	% self-initiated corrections	0%	60%	Best documented CRA self-correction rate; ceiling conservative
AC	% audit recs implemented	20%	100%	Worst documented audit compliance period
CRR _c	% complaints resolved	25%	95%	NCRAAs rarely achieve 100% consumer-defined resolution
RAR	% remediation on schedule	10%	100%	Consent order compliance data; floor = documented non-compliance

Data Source Credibility Hierarchy

Conflicting quanta from multiple sources are resolved by the following tier weights:

Tier	Weight	Source Description
1	1.00	Court orders, consent decrees, regulatory enforcement actions signed by judicial or regulatory authority
2	0.90	IG reports, GAO findings, CFPB examination summaries, congressional testimony under oath
3	0.75	CFPB complaint database (aggregated), FTC Consumer Sentinel, State AG formal complaints
4	0.50	SEC/LSE public filings (self-reported — cross-validation required)
5	0.20	Third-party research studies, academic analyses, news reporting (corroboration required)
6	0.05	Individual platform submissions — supplementary signal only, never primary evidence

MongoDB Integration Note: MongoDB is well-suited to store Tier 5–6 unstructured quanta (research PDFs, scraped records, individual submissions) before normalization into the canonical Postgres `behavioral_quanta` table. Recommended pattern: raw ingestion → MongoDB document store → normalization pipeline → Postgres canonical storage → Neo4j graph relationships. MongoDB serves as the landing zone for heterogeneous source types before the Signal Harmonizer Agent processes them into typed quanta.

Implementation Build Order — Beacon Pipeline

Step	Deliverable	Description	Timeline
1	Behavioral Quanta Schema	Deploy <code>behavioral_quanta</code> and <code>field_interactions</code> tables with <code>source_type</code> enum, <code>credibility_tier</code> , and <code>magnitude</code> columns	Day 1–2
2	CRA Data Connectors	CFPB complaint connector, PACER/CourtListener, SEC EDGAR, FTC docket. Each emits tier-tagged quanta. MongoDB landing zone for raw documents.	Day 3–10
3	Normalization Service	Min-max with CRA bounds table; configurable per institution class; clamp to $[0, 1]$; store in <code>behavioral_quanta.magnitude</code>	Day 11–13
4	Six Dimensional Agents	Build agent pairs in order: Fidelity → Consistency → Transparency → Efficiency → Equity → Accountability	Day 14–35
5	Hard Floor Enforcement	Equity and Accountability hard floor logic; must fire before composite computation; log all applications	Day 30–33
6	Composite Score	$A = \sum w_i \cdot D_i$ with AHP weights; configurable per institution class; store all dimensional intermediates	Day 36–38
7	Tier Assignment	Apply thresholds; trigger tier-specific workflows (flag, alert, referral)	Day 39–40
8	CRA Pilot Scoring	Full Aptitude Score for all three NCRAAs using historical public data; validate against known enforcement history	Day 41–50
9	Reproducibility Audit	Re-derive score from raw quanta; verify byte-level match; document audit trail	Day 51–53
10	Public Score API	Expose scores with full dimension breakdown, quantum count, source citations, confidence intervals	Day 54–60

Target: Working CRA Aptitude Scores within 60 days of pipeline build start.